

Data-Driven Short-Term Grid Frequency Stability Forecast

Final Code Output Graphs - GitHub Ready Visual Report

This PDF compiles the final generated graph outputs from the Python forecasting pipeline. The sample dataset is synthetic and reproducible, while the workflow is ready to be extended with ENTSO-E-compatible frequency data and NASA POWER weather features.

Metric	Value
MAE	0.02040
RMSE	0.02546
R2	-0.001
Reliability within +/-10 mHz	30.1%

Figure 1. Grid Frequency Time Series

Shows the simulated short-term frequency variation around the nominal 50 Hz reference line.

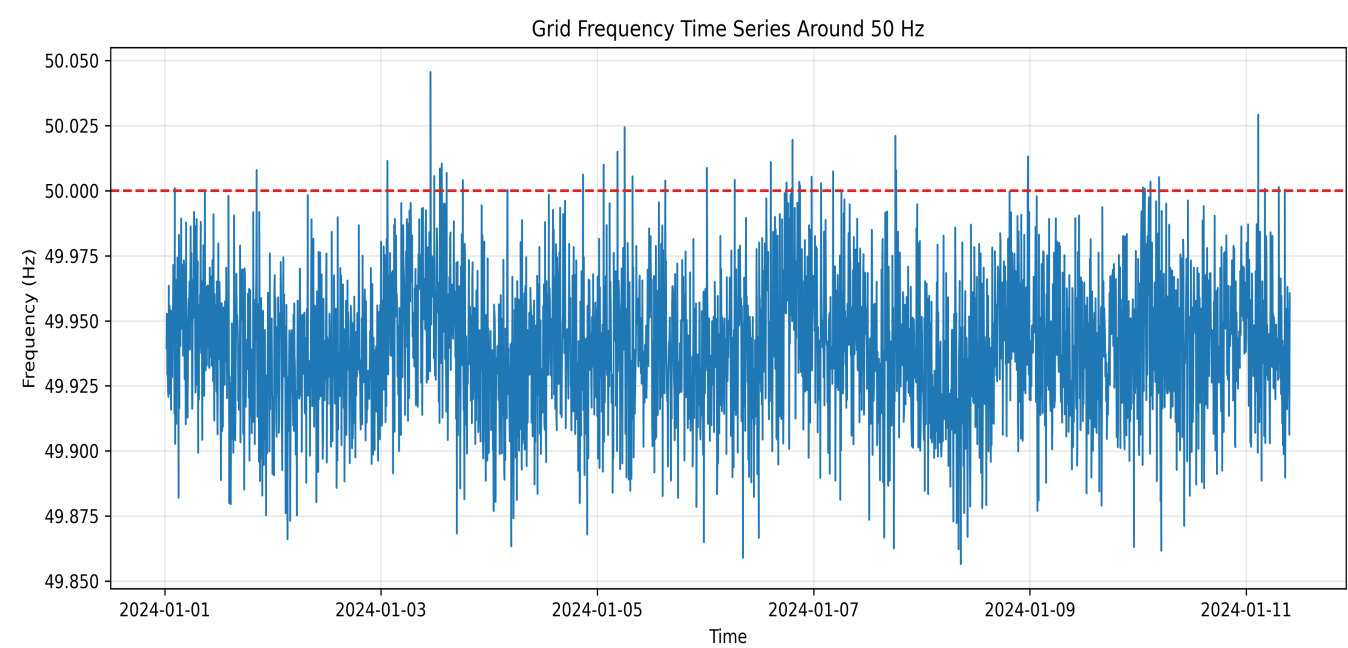


Figure 2. Actual vs Predicted Forecast

Compares 15-minute-ahead true deviation with Random Forest prediction.

15-Minute Ahead Actual vs Predicted Frequency Deviation

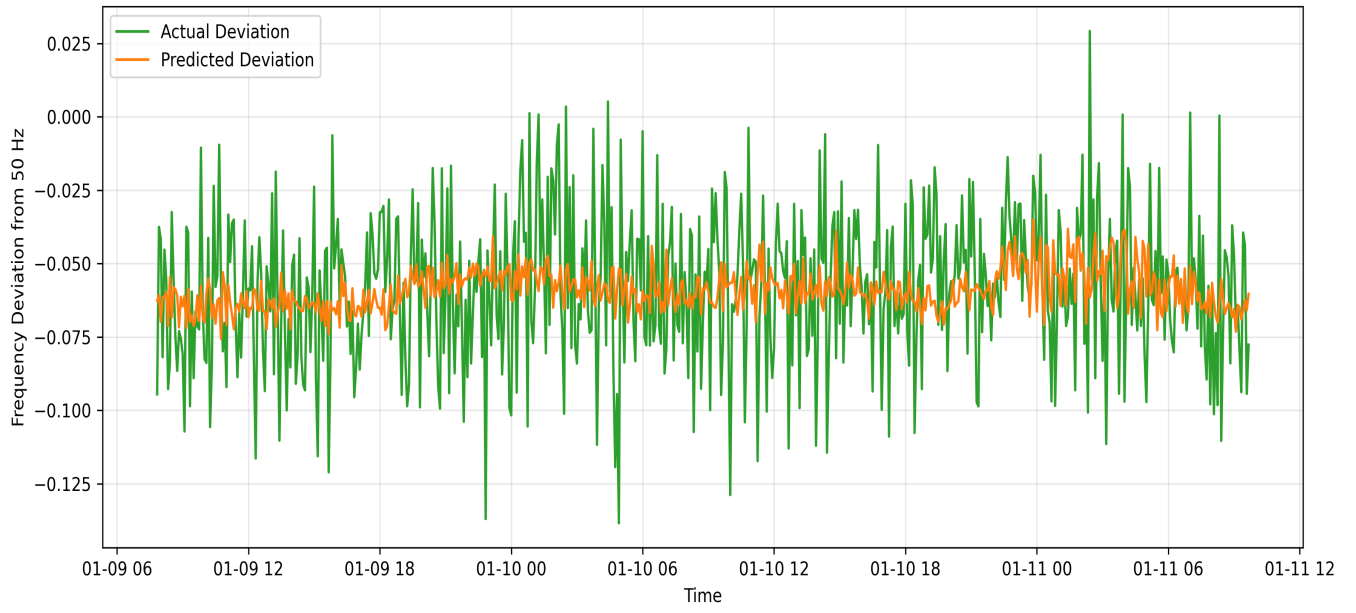


Figure 3. Residual Error Distribution

Displays the model error spread for reliability and bias analysis.

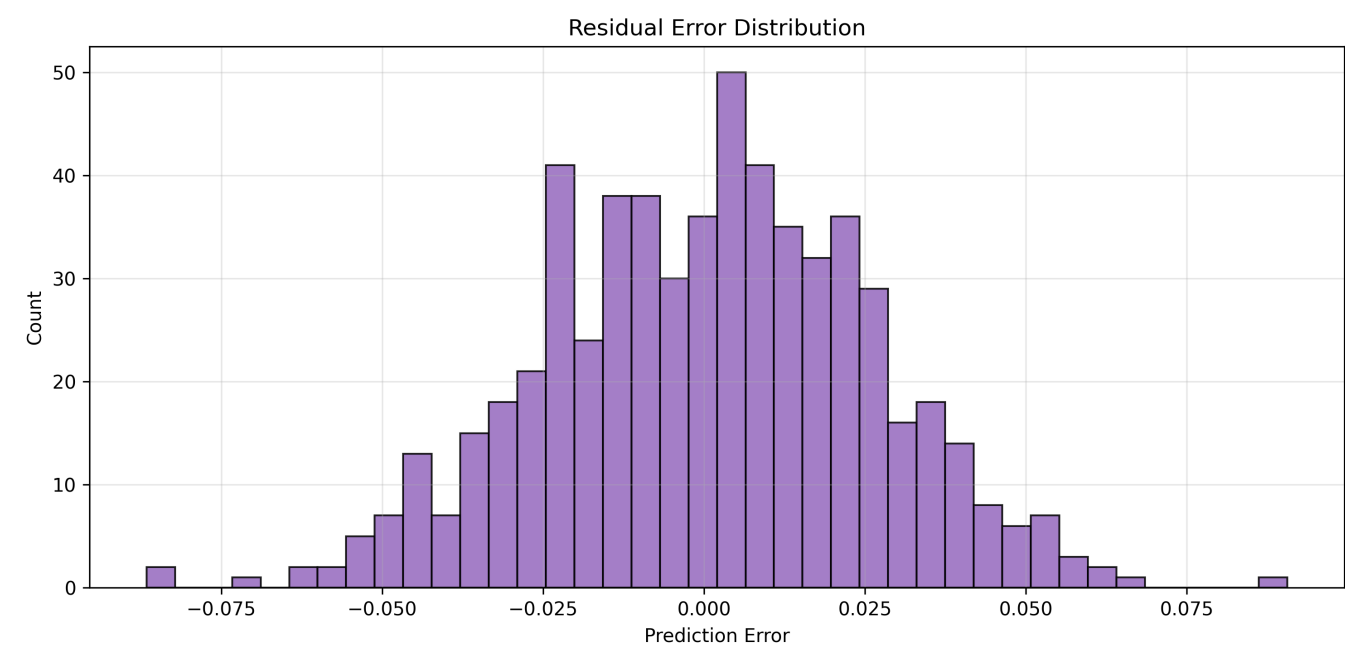


Figure 4. Feature Importance

Ranks weather, renewable proxy, load, and lagged frequency features by predictive contribution.

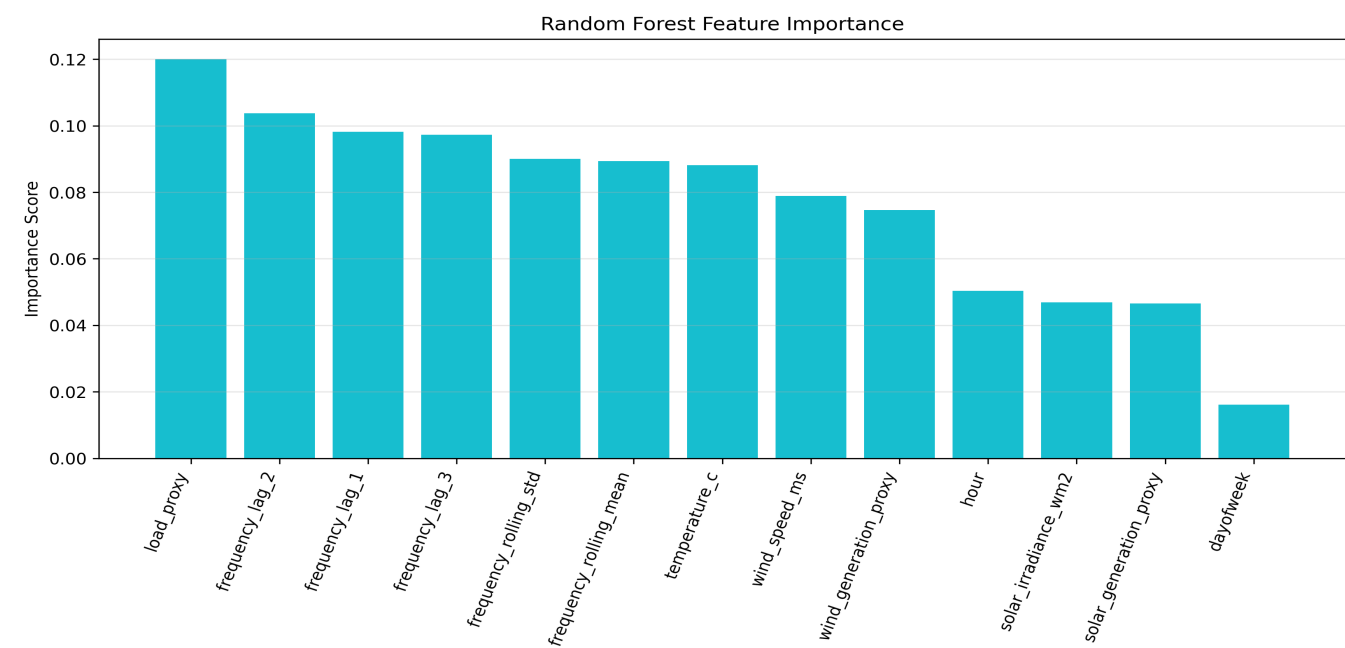


Figure 5. Renewable Generation Proxies

Visualizes solar and wind proxy behavior used for renewable-aware forecasting.

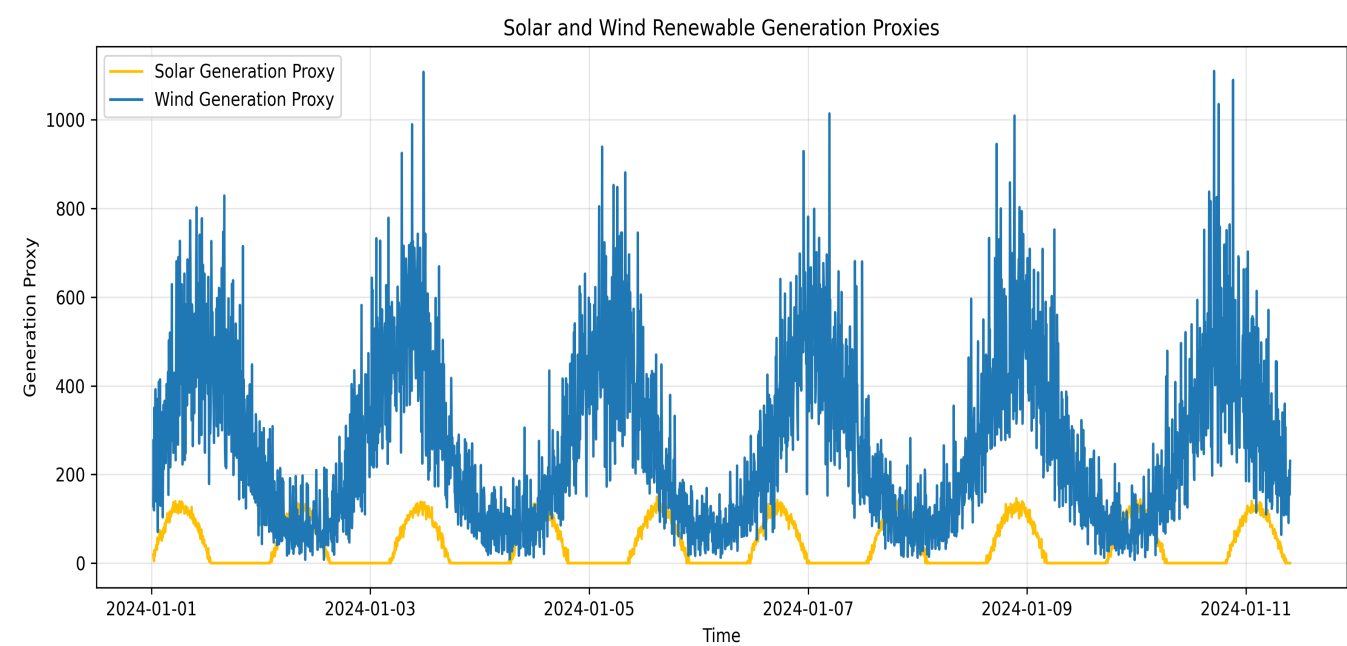


Figure 6. Load vs Renewables

Compares load proxy with solar and wind proxies to indicate balancing stress.

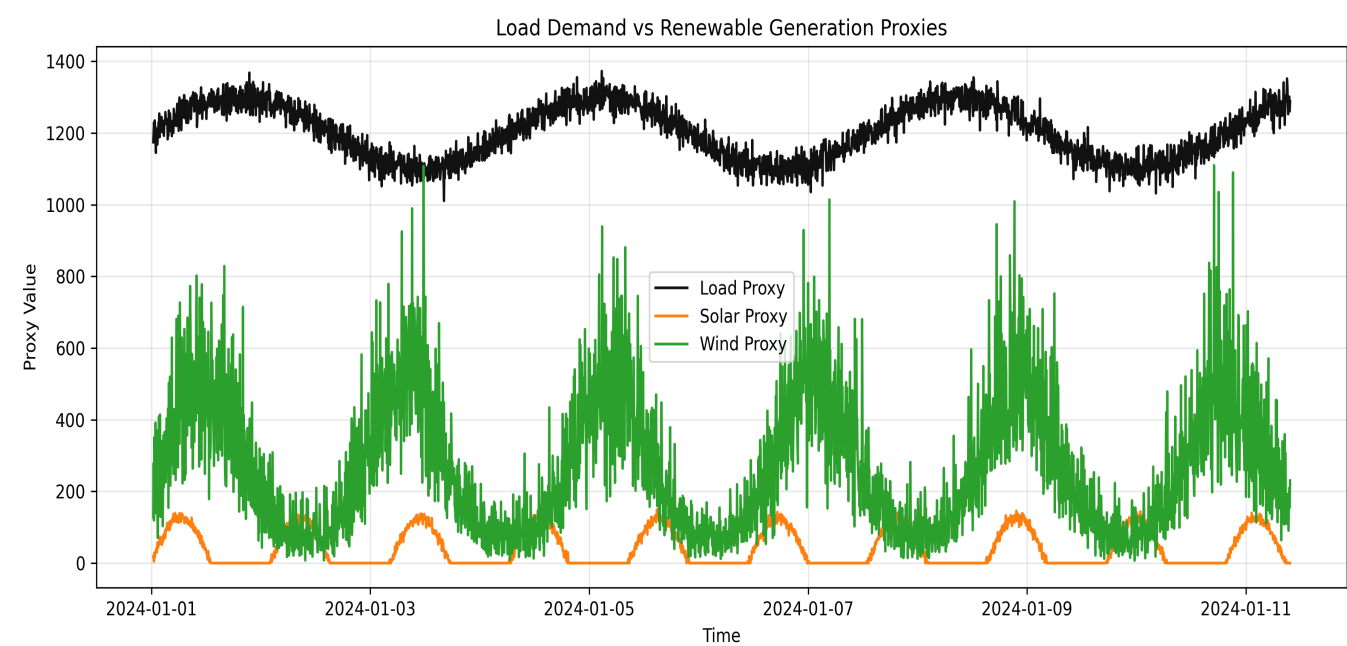


Figure 7. Actual vs Predicted Scatter

Shows point-wise agreement between measured and predicted deviation values.

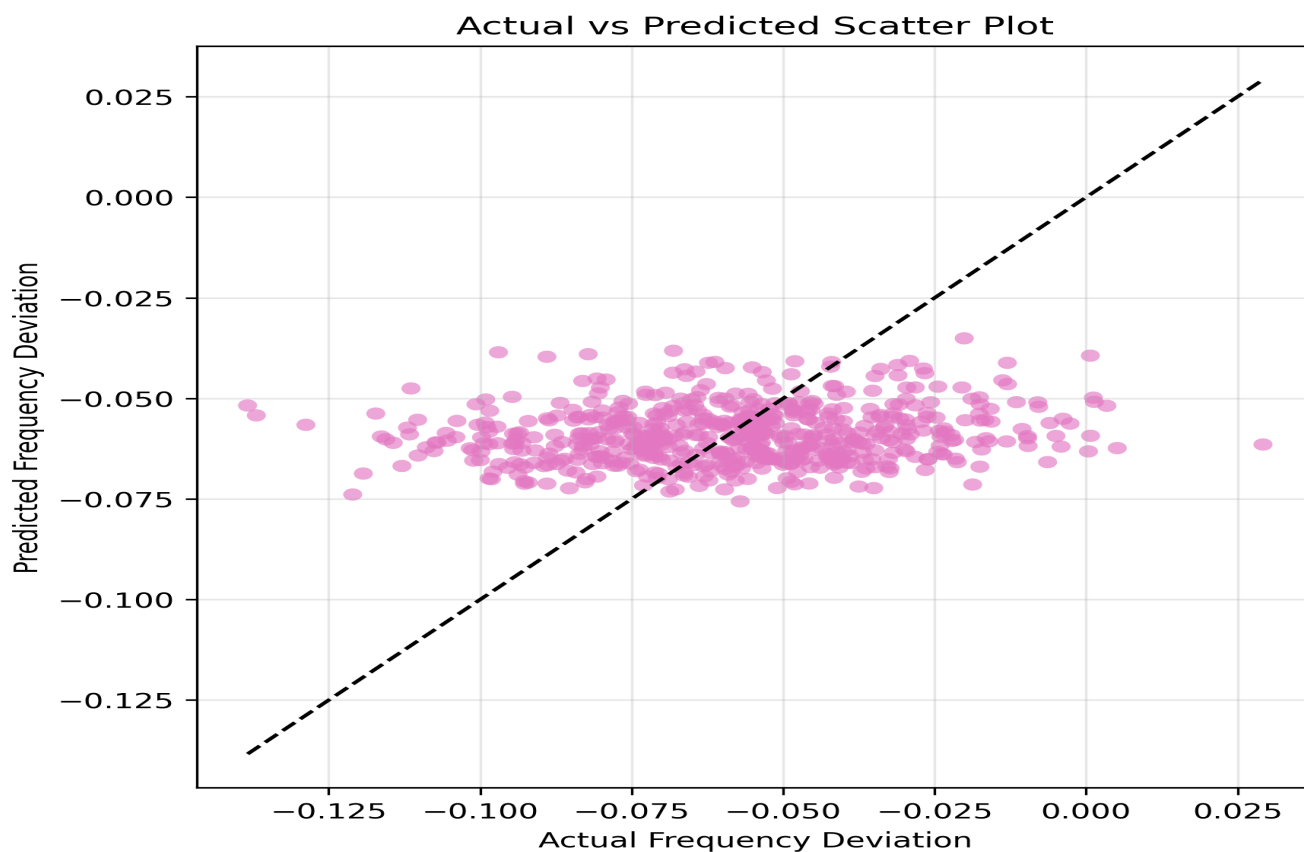


Figure 8. Model Performance Metrics

Summarizes MAE and RMSE values for the final Random Forest pipeline.

